IDEABATIC

SMILE GO Project

Uniform cooling solutions

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Problem Overview

Reference: https://www.unicef.org/supply/vaccine-carriers

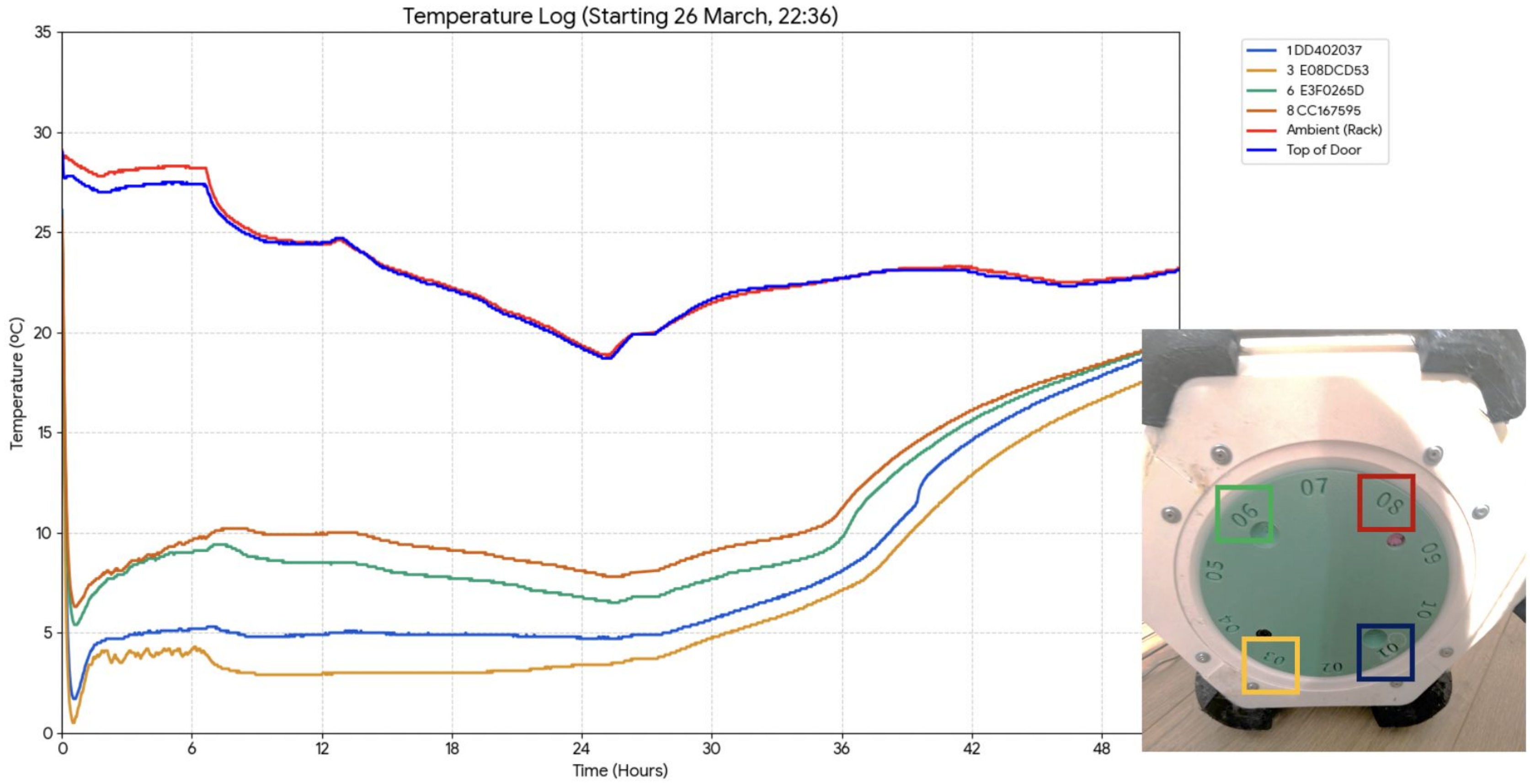
Problem Overview

Problem: Asymmetric cooling

- ➢ Warmer vaccines at the top, cooler vaccines at the bottom
- ➢ Conduction & Convection

Current prototype of SMILE GO:

- Sized down SMILE
- No asymmetric cooling mitigation in the bottle
- Foam spray insulation

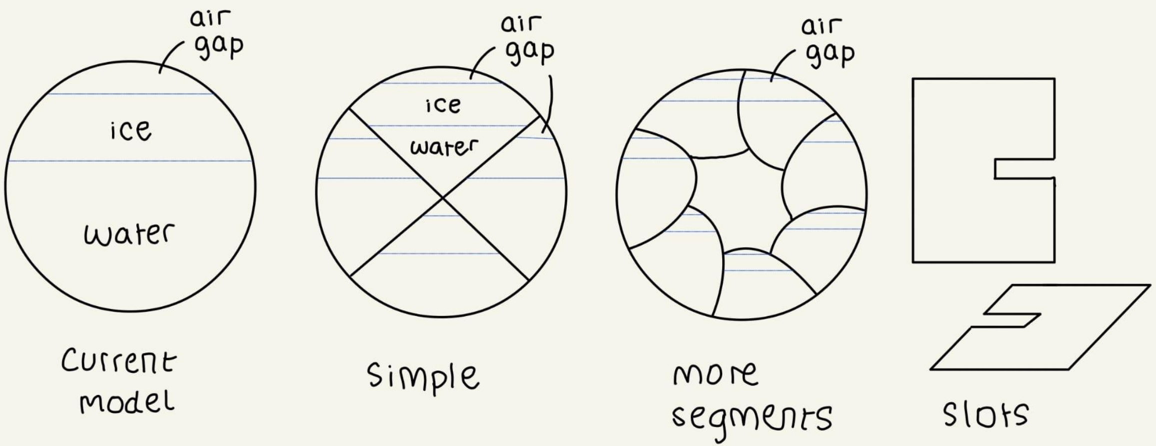


Temperature Log (Starting 26 March, 22:36)
Temperature (ºC)
Time (Hours)
35
30
25
20
15
10
5
0
0
6
12
18
24
30
36
42
48
1 DD402037
3 E08DCD53
6 E3F0265D
8 CC167595
Ambient (Rack)
Top of Door
07
06
08
05
09
04
10
03
02
01

Proposal

Investigate causes and ways to reduce asymmetric cooling

- Design & test cooler inserts (for example on the right)
    - Trade off of increasing insert volume and decreasing the water content
- Investigate impact of convection
- Prototype new carousel design to reduce convection



Temperature sensing the current model, the inserts and the convection carousel to compare

Proposal Qualities

- **Responsible** → no chemicals, no damage, recyclable
- **Responsive** → proposal developed in consultation with partner
- **Adds significant value** → increase vaccine cool-life
- **Accessible to end users** → modifications don’t affect use
- **Context appropriate** → higher ambient temperature, 3d printable, accessible materials

Aligns with UN sustainable goals:

Skills

Karen

Strengths:
- 3D modelling
- 3D printing
- Drawing

Weaknesses:
- Thermal modelling
- Hardware

Kerry

Strengths:
- Thermal/Material knowledge
- Mechanical Design

Weaknesses:
- Computational modelling
- Electronics

Kavita

Strengths:
- Electronics
- 3D printing experience

Weaknesses:
- Computational modelling

Possible Training Required:
- Laser Cutting
- Soldering

Skills required for the project:
- 3D printing
- Prototyping
- Data analysis
- Time management

Project Development Timeline

	TASK TITLE	18/05 - 24/05							25/05 - 31/05							1/06 - 7/06							8/06 - 11/06			
		M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T
	Proposal Stage																									
	Project Planning																									
	Meeting with Partner																							C		
	Work on proposal presentation																							O		
	Proposal Presentation																							N		
	Testing & Prototyping																							T		
Control Test	Sort electronics testing equipment																							I		
	Control test on current prototype								Test															N		
Cooler Insert	Research & design initial insert																							G		
	Print inserts																							E		
	Test various insert designs														Test 1		Test 2		Test 3					N		
	Analyse insert results																							C		
	Work on interim presentation																							Y		
	Interim Presentation																									
Convection	Convection tests										Test 1		Test 2											D		
	Convection design prototype																							A		
	Print & develop carousel prototype																							Y		
	Test Prototype																				Test			S		
	1-to-1 meetings																									
	Final Stage																									
	Work on presentations/report																									
	Final submissions																									

	If alternate plan needed											Underfilling bottle, different orientation tests etc														

Contingency Plan

- **Extra Time:** Contingency days at end to account for testing taking longer than expected
- **Staged testing approach:** test out concepts before we put significant time into designing
- **Other ideas:** Some alternatives areas to look at if our initial testing suggests the concept is not viable e.g. effect of underfilling ice bottle (does this exacerbate asymmetric cooling), insulation methods & materials
- **Reduce risks:** complete Risk Assessments

Risk Assessment

Tooling

- Effect - Burns from hot glue guns, soldering burns, breathing in toxic fumes, fire from laser cutter
- Mitigation - acquire appropriate training, make sure fume hood on laser cutter is activated.

Electrical Testing

- Effect - water contact with electrical components, overheating of Arduino.
- Mitigation - keep electronics away from water, keep surfaces dry, and keep voltages within working range.

3D Printer

- Effect - cuts from sharp components from 3D printer, high temperatures on 3D printer nozzle
- Mitigation - distance from 3D printer as it is operating, use non-hazardous materials in 3D printer.

Resources

Resource	Cost	Source
PLA printer filament	£10-15	CUED
Ice Bottle	£10	RS components
Temperature Sensors	£25	Amazon
Arduino Wires	£0	CUED
Laser Cut Wood (Offcuts)	£0	CUED
Hot Glue Sticks	£1	CUED

Thank You for Listening!

Any Questions?